

Probability Rules & Conditional Probability



Basic probability rules

- 1 A fair six-sided die is rolled. Find the probability the roll is even, and the probability the roll is greater than 4.
- 2 For events A and B , $P(A) = 0.4$, $P(B) = 0.3$, and $P(A \cup B) = 0.6$. Find $P(A \cap B)$ using the addition rule.

Mutually exclusive and independent events

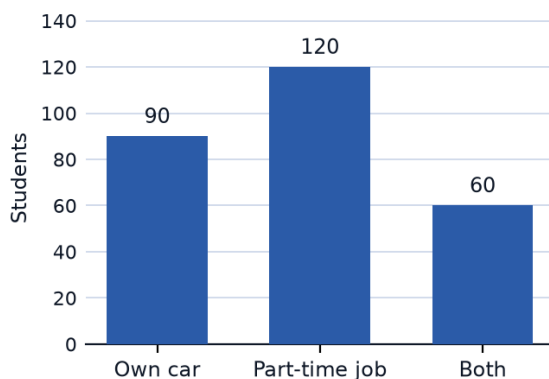
- 3 Explain the difference between mutually exclusive events and independent events, and give a brief example of each.
- 4 Two events A and B satisfy $P(A) = 0.5$ and $P(B) = 0.5$, and they are mutually exclusive. Explain why A and B cannot also be independent (unless one has probability 0).

Conditional probability

- 5 In a class, the probability a student likes math is 0.6, and the probability a student likes both math and science is 0.42. Find the probability a student likes science given that they like math.
- 6 Use the multiplication rule to find $P(A \cap B)$ if $P(A) = 0.5$ and $P(B | A) = 0.4$.

Two-way tables and conditional probability

- 7 A survey of 200 students finds 90 own a car, 120 have a part-time job, and 60 both own a car and have a job. Find the probability a randomly selected student has a job, given that they own a car.



- 8 Using the same survey data, determine whether owning a car and having a job are independent events, showing your reasoning.
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Tree diagrams and sequential events

- 9 A factory has two machines: Machine A produces 60% of items with a 2% defect rate, and Machine B produces 40% of items with a 5% defect rate. Find the probability a randomly selected item is defective, using the law of total probability.
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Applying probability rules to real contexts

- 10 A weather forecaster states there is a 30% chance of rain on Saturday and a 30% chance of rain on Sunday, and treats the two days as independent. Find the probability it rains on both days, and the probability it rains on at least one of the two days.
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The complement rule

- 11 A box of 50 light bulbs contains 6 defective bulbs. If one bulb is selected at random, find the probability it is NOT defective, using the complement rule.
- 12 A student estimates there is a 65% chance of rain at least once during a 3-day trip, assuming each day is independent with the same daily rain probability p . Find p , to three decimal places, using $P(\text{no rain all 3 days}) = (1 - p)^3$.
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Combining independence and conditional probability

- 13 Two lie detector tests are given to a suspect, each correctly detecting deception 90% of the time (independently of each other). If the suspect is actually lying, find the probability both tests correctly detect it.
- 14 A bag contains 5 red marbles and 3 blue marbles. Two marbles are drawn without replacement. Find the probability both are red.
- 15 At a certain hospital, 40% of patients admitted have condition A, and 25% have condition B. Additionally, 10% of all patients have both conditions A and B. Draw on your understanding of the addition rule to find the probability a randomly selected patient has condition A or condition B (or both), and separately explain why simply adding $P(A) + P(B)$ without adjustment would overcount certain patients.

- 16 An airline reports that 85% of its flights depart on time. Among on-time flights, 95% also arrive on time, while among delayed departures, only 40% arrive on time. Using a tree-diagram approach and the law of total probability, find the overall probability that a randomly selected flight arrives on time.
- 17 A quality inspector at a factory knows that 3% of all items produced are defective. The inspection process correctly identifies a defective item 98% of the time, but also incorrectly flags a non-defective item as defective 2% of the time (a false positive). If an item is flagged as defective by the inspection process, use the given information and the multiplication rule to find the probability that the item actually IS defective, showing every step of your reasoning (this requires combining the law of total probability with conditional probability, similar to Bayes' theorem).
- 18 A survey of 500 adults finds that 280 exercise regularly, 220 eat a balanced diet, and 150 do both. Using this information, first find the probability a randomly selected adult exercises regularly or eats a balanced diet (or both), and then find the probability an adult eats a balanced diet given that they exercise regularly, showing every step of your work clearly.
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Multiple choice — easy 15

- 19 For events A and B , $P(A) = 0.30$, $P(B) = 0.70$, and $P(A \cap B) = 0.05$. Find $P(A \cup B)$ using the addition rule.
- a. 0.90
 - b. 1.05
 - c. 0.95
 - d. 1.00
- 20 For events A and B , $P(A) = 0.50$, $P(B) = 0.55$, and $P(A \cap B) = 0.25$. Find $P(A \cup B)$ using the addition rule.
- a. 1.05
 - b. 0.80
 - c. 0.85
 - d. 0.75

- 21 For events A and B , $P(A) = 0.35$, $P(B) = 0.60$, and $P(A \cap B) = 0.25$. Find $P(A \cup B)$ using the addition rule.
- a. 0.70
 - b. 0.95
 - c. 0.75
 - d. 0.65
- 22 For events A and B , $P(A) = 0.40$, $P(B) = 0.35$, and $P(A \cap B) = 0.35$. Find $P(A \cup B)$ using the addition rule.
- a. 0.45
 - b. 0.75
 - c. 0.40
 - d. 0.35
- 23 For events A and B , $P(A) = 0.25$, $P(B) = 0.55$, and $P(A \cap B) = 0.20$. Find $P(A \cup B)$ using the addition rule.
- a. 0.55
 - b. 0.80
 - c. 0.65
 - d. 0.60
- 24 For events A and B , $P(A) = 0.70$, $P(B) = 0.30$, and $P(A \cap B) = 0.05$. Find $P(A \cup B)$ using the addition rule.
- a. 1.00
 - b. 1.05
 - c. 0.90
 - d. 0.95

- 25 For events A and B , $P(A) = 0.35$, $P(B) = 0.50$, and $P(A \cap B) = 0.10$. Find $P(A \cup B)$ using the addition rule.
- a. 0.75
 - b. 0.80
 - c. 0.70
 - d. 0.85
- 26 For events A and B , $P(A) = 0.60$, $P(B) = 0.65$, and $P(A \cap B) = 0.55$. Find $P(A \cup B)$ using the addition rule.
- a. 0.75
 - b. 0.70
 - c. 1.25
 - d. 0.65
- 27 For events A and B , $P(A) = 0.40$, $P(B) = 0.60$, and $P(A \cap B) = 0.40$. Find $P(A \cup B)$ using the addition rule.
- a. 0.60
 - b. 1.00
 - c. 0.65
 - d. 0.55
- 28 For events A and B , $P(A) = 0.35$, $P(B) = 0.55$, and $P(A \cap B) = 0.30$. Find $P(A \cup B)$ using the addition rule.
- a. 0.65
 - b. 0.90
 - c. 0.55
 - d. 0.60

- 29 For events A and B , $P(A) = 0.55$, $P(B) = 0.35$, and $P(A \cap B) = 0.25$. Find $P(A \cup B)$ using the addition rule.
- a. 0.60
 - b. 0.70
 - c. 0.90
 - d. 0.65
- 30 For events A and B , $P(A) = 0.60$, $P(B) = 0.65$, and $P(A \cap B) = 0.25$. Find $P(A \cup B)$ using the addition rule.
- a. 0.95
 - b. 1.00
 - c. 1.05
 - d. 1.25
- 31 For events A and B , $P(A) = 0.20$, $P(B) = 0.60$, and $P(A \cap B) = 0.20$. Find $P(A \cup B)$ using the addition rule.
- a. 0.60
 - b. 0.65
 - c. 0.55
 - d. 0.80
- 32 For events A and B , $P(A) = 0.55$, $P(B) = 0.45$, and $P(A \cap B) = 0.45$. Find $P(A \cup B)$ using the addition rule.
- a. 0.50
 - b. 0.55
 - c. 0.60
 - d. 1.00

- 33 For events A and B , $P(A) = 0.20$, $P(B) = 0.60$, and $P(A \cap B) = 0.10$. Find $P(A \cup B)$ using the addition rule.
- a. 0.75
 - b. 0.65
 - c. 0.80
 - d. 0.70
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Multiple choice — medium 15

- 34 For events A and B , $P(A) = 0.30$ and $P(A \cap B) = 0.30$. Find $P(B \mid A)$.
- a. 1.10
 - b. 1.00
 - c. 0.90
 - d. 0.00
- 35 For events A and B , $P(A) = 0.50$ and $P(A \cap B) = 0.40$. Find $P(B \mid A)$.
- a. 0.10
 - b. 0.90
 - c. 0.80
 - d. 0.70
- 36 For events A and B , $P(A) = 0.80$ and $P(A \cap B) = 0.25$. Find $P(B \mid A)$.
- a. 0.41
 - b. 0.21
 - c. 0.31
 - d. 0.55

- 37 For events A and B , $P(A) = 0.30$ and $P(A \cap B) = 0.10$. Find $P(B \mid A)$.
- a. 0.43
 - b. 0.20
 - c. 0.33
 - d. 0.23
- 38 For events A and B , $P(A) = 0.45$ and $P(A \cap B) = 0.10$. Find $P(B \mid A)$.
- a. 0.12
 - b. 0.35
 - c. 0.22
 - d. 0.32
- 39 For events A and B , $P(A) = 0.80$ and $P(A \cap B) = 0.40$. Find $P(B \mid A)$.
- a. 0.60
 - b. 0.50
 - c. 0.40
 - d. 0.55
- 40 For events A and B , $P(A) = 0.40$ and $P(A \cap B) = 0.05$. Find $P(B \mid A)$.
- a. 0.03
 - b. 0.35
 - c. 0.23
 - d. 0.12

- 41 For events A and B , $P(A) = 0.65$ and $P(A \cap B) = 0.60$. Find $P(B | A)$.
- a. 0.92
 - b. 1.02
 - c. 0.82
 - d. 0.05
- 42 For events A and B , $P(A) = 0.55$ and $P(A \cap B) = 0.40$. Find $P(B | A)$.
- a. 0.83
 - b. 0.73
 - c. 0.63
 - d. 0.15
- 43 For events A and B , $P(A) = 0.30$ and $P(A \cap B) = 0.15$. Find $P(B | A)$.
- a. 0.15
 - b. 0.60
 - c. 0.40
 - d. 0.50
- 44 For events A and B , $P(A) = 0.70$ and $P(A \cap B) = 0.60$. Find $P(B | A)$.
- a. 0.10
 - b. 0.86
 - c. 0.76
 - d. 0.96

- 45 For events A and B , $P(A) = 0.50$ and $P(A \cap B) = 0.50$. Find $P(B \mid A)$.
- a. 0.00
 - b. 1.10
 - c. 0.90
 - d. 1.00
- 46 For events A and B , $P(A) = 0.70$ and $P(A \cap B) = 0.35$. Find $P(B \mid A)$.
- a. 0.50
 - b. 0.35
 - c. 0.60
 - d. 0.40
- 47 For events A and B , $P(A) = 0.85$ and $P(A \cap B) = 0.10$. Find $P(B \mid A)$.
- a. 0.75
 - b. 0.22
 - c. 0.12
 - d. 0.02
- 48 For events A and B , $P(A) = 0.35$ and $P(A \cap B) = 0.10$. Find $P(B \mid A)$.
- a. 0.29
 - b. 0.39
 - c. 0.25
 - d. 0.19

Multiple choice — hard 15

- 49 A bag contains 7 red marbles and 9 blue marbles. Two marbles are drawn without replacement. Find the probability both are red.
- a. $\frac{7}{40}$
 - b. $\frac{11}{40}$
 - c. $\frac{3}{40}$
 - d. $\frac{7}{16}$
- 50 A bag contains 7 red marbles and 6 blue marbles. Two marbles are drawn without replacement. Find the probability both are blue.
- a. $\frac{19}{65}$
 - b. $\frac{6}{65}$
 - c. $\frac{6}{13}$
 - d. $\frac{5}{26}$
- 51 A bag contains 4 red marbles and 6 blue marbles. Two marbles are drawn without replacement. Find the probability both are blue.
- a. $\frac{1}{3}$
 - b. $\frac{3}{5}$
 - c. $\frac{13}{30}$
 - d. $\frac{7}{30}$
- 52 A bag contains 6 red marbles and 4 blue marbles. Two marbles are drawn without replacement. Find the probability both are red.
- a. $\frac{7}{30}$
 - b. $\frac{13}{30}$
 - c. $\frac{1}{3}$
 - d. $\frac{3}{5}$

- 53 A bag contains 4 red marbles and 8 blue marbles. Two marbles are drawn without replacement. Find the probability both are blue.
- a. $\frac{107}{330}$
 - b. $\frac{14}{33}$
 - c. $\frac{2}{3}$
 - d. $\frac{173}{330}$
- 54 A factory has two machines: Machine A produces 60% of items with a 10% defect rate, and Machine B produces 40% of items with a 15% defect rate. Letting D denote the event that an item is defective, find $P(D)$ for a randomly selected item, using the law of total probability.
- a. 12.00%
 - b. 13.00%
 - c. 11.50%
 - d. 12.50%
- 55 A factory has two machines: Machine A produces 80% of items with a 10% defect rate, and Machine B produces 20% of items with a 3% defect rate. Letting D denote the event that an item is defective, find $P(D)$ for a randomly selected item, using the law of total probability.
- a. 8.60%
 - b. 4.40%
 - c. 8.10%
 - d. 9.10%

- 56 A factory has two machines: Machine A produces 40% of items with a 4% defect rate, and Machine B produces 60% of items with a 1% defect rate. Letting D denote the event that an item is defective, find $P(D)$ for a randomly selected item, using the law of total probability.
- a. 1.70%
 - b. 2.20%
 - c. 2.80%
 - d. 2.70%
- 57 A factory has two machines: Machine A produces 50% of items with a 9% defect rate, and Machine B produces 50% of items with a 2% defect rate. Letting D denote the event that an item is defective, find $P(D)$ for a randomly selected item, using the law of total probability.
- a. 6.00%
 - b. 5.50%
 - c. 6.50%
 - d. 5.00%
- 58 A factory has two machines: Machine A produces 60% of items with a 5% defect rate, and Machine B produces 40% of items with a 11% defect rate. Letting D denote the event that an item is defective, find $P(D)$ for a randomly selected item, using the law of total probability.
- a. 6.90%
 - b. 7.90%
 - c. 8.60%
 - d. 7.40%

- 59 A quality inspector knows 6% of all items produced are defective (event D). The inspection process correctly flags a defective item 85% of the time, but also flags a non-defective item as defective 1% of the time (a false positive). If an item is flagged as defective (event F), find $P(D | F)$, to one decimal place.
- a. 85.0%
 - b. 81.4%
 - c. 87.4%
 - d. 84.4%
- 60 A quality inspector knows 5% of all items produced are defective (event D). The inspection process correctly flags a defective item 93% of the time, but also flags a non-defective item as defective 6% of the time (a false positive). If an item is flagged as defective (event F), find $P(D | F)$, to one decimal place.
- a. 93.0%
 - b. 47.9%
 - c. 44.9%
 - d. 41.9%
- 61 A quality inspector knows 8% of all items produced are defective (event D). The inspection process correctly flags a defective item 96% of the time, but also flags a non-defective item as defective 4% of the time (a false positive). If an item is flagged as defective (event F), find $P(D | F)$, to one decimal place.
- a. 96.0%
 - b. 70.6%
 - c. 64.6%
 - d. 67.6%

- 62** A quality inspector knows 3% of all items produced are defective (event D). The inspection process correctly flags a defective item 97% of the time, but also flags a non-defective item as defective 6% of the time (a false positive). If an item is flagged as defective (event F), find $P(D | F)$, to one decimal place.
- a. 30.3%
 - b. 97.0%
 - c. 33.3%
 - d. 36.3%
- 63** A quality inspector knows 4% of all items produced are defective (event D). The inspection process correctly flags a defective item 99% of the time, but also flags a non-defective item as defective 1% of the time (a false positive). If an item is flagged as defective (event F), find $P(D | F)$, to one decimal place.
- a. 83.5%
 - b. 77.5%
 - c. 80.5%
 - d. 99.0%